1. Introduction

Evaluating Large Language Models (LLMs) in the general domain primarily focuses on fluency, coherence, engagement, and broad stylistic appeal. In healthcare, however, these criteria are insufficient. A general-purpose language model can generate an output that is grammatically flawless, highly empathetic, and syntactically persuasive, yet medically inaccurate or catastrophically unsafe. Because clinical decisions directly impact patient morbidity and mortality, evaluating a Medical LLM requires shifting from traditional natural language processing (NLP) metrics toward strict clinical correctness, factual precision, and safety guarantees.

2. Research Papers Reviewed

To establish a baseline of current evaluation methodologies, five key research papers across medical AI evaluation, benchmark datasets, and clinical alignment were analyzed.

i. Large language models encode clinical knowledge

➔ Evaluates PaLM, Flan-PaLM, and Med-PaLM on Medical QA tasks using the MultiMedQA benchmark (3,173 consumer questions plus standard datasets). It combines automated MCQ accuracy with a 12-axis physician rubric evaluation measuring scientific consensus alignment, potential harm, and omission rates. Key results showed Flan-PaLM achieved 67.6% accuracy on MedQA, while Med-PaLM instruction tuning matched physician baselines on scientific consensus (92.9%).

ii. Toward expert-level medical question answering with large language models

➔ Evaluates Med-PaLM 2 across expert-level medical QA and consultation support tasks using 1,061 USMLE queries, consumer questions, and adversarial datasets. The evaluation framework uses pairwise clinician preference ranking and a physician safety audit measuring accuracy, preference win-rate, and harm potential. Med-PaLM 2 achieved 86.5% on MedQA and was preferred over generalist physician answers 65% of the time, though subtle hallucinations remained in complex subspecialties.

iii. Towards evaluating and building versatile large language models for medicine

➔ Studies versatile clinical NLP models (MMedIns-Llama 3, MEDITRON, Llama 3, GPT-4, Claude-3.5) across 52 clinical tasks on the MedS-Bench benchmark. Evaluation relies on automated string metrics (Exact Match, BLEU-1/4, ROUGE-L), BERTScore, and radiologist/clinician Likert ratings. It highlighted that while models perform well on multiple-choice questions, standard automated NLP metrics correlate poorly with clinical safety and diagnostic validity on non-MCQ tasks

iv. MedGPTEval: A Dataset and Benchmark to Evaluate Responses of LLMs in Medicine

➔ Assesses ChatGPT, ERNIE Bot, and Doctor PuJiang on multi-turn medical dialogues and clinical case report generation across 34 clinical scenario contexts. Evaluated by five licensed medical experts using a Delphi-optimized 16-indicator rubric focusing on professional capability, empathy, contextual coherence, and robustness. Results demonstrated that the domain-tuned Doctor PuJiang model surpassed general LLMs in multi-turn coherence and empathy.

v. MedHallu: A Benchmark for Medical Hallucination Detection

➔ Focuses on medical hallucination detection in models like GPT-4o, Llama-3.1, UltraMedical, BioMistral, and Qwen2.5 using 10,000 QA pairs derived from PubMedQA across four hallucination categories. Evaluates performance using precision, recall, macro/micro F1-score, and abstention rate via automated bidirectional entailment verification. Findings revealed that SOTA models struggle on hard hallucinations (F1 as low as 0.625), but introducing an explicit "Not Sure" abstention option improved precision by 38%.

3. Evaluation Methods Found

Research reveals five primary paradigms currently used to evaluate Medical LLMs:

i. **Automated Lexical Overlap:** Measures exact surface-level token matching between model outputs and reference answers (n-gram overlaps like BLEU and ROUGE).

ii. **Automated Semantic Embeddings:** Measures contextual vector similarity using specialized encoders (e.g., BERTScore using BioBERT embeddings)

iii. **Factuality & LLM-as-a-Judge Frameworks:** Utilizes advanced LLMs (e.g., GPT-4) prompted with clinical rubrics to extract claims into atomic facts and verify each claim against medical guidelines.

iv. **Physician & Expert Rubric Evaluation:** Qualitative assessment performed by practicing clinicians scoring outputs across multidimensional axes: factual accuracy, severity of potential harm, completeness, and clinical reasoning validity.

v. **Preference & Pairwise Alignment:** Clinicians or automated judges evaluate side-by-side outputs from two models to establish win-rates and Bradley-Terry preference scores.

4. Metrics Identified

The following 10 evaluation metrics and methods were synthesized from the research literature:

Metric Name	Metric Category	Formula / Scoring Logic	Primary Evaluation Focus	Advantages	Limitations / Clinical Risks	Sourc Paper
MedQA Accuracy	Objective Benchmark	Accuracy = Correct / Total	Multiple-choice medical exam knowledge retrieval.	100% objective, instantly automated, standard across benchmarks.	Tests test-taking memory, not clinical reasoning or conversational safety.	Paper 1, Paper 2, Paper 3
xact Match EM)	String Matching	EM = 1 if Pred == Ref else 0	Absolute string identity for clinical entity extraction.	Strict and binary; ideal for ICD-10 codes, lab values, and drug names.	Overly rigid; penalizes valid synonyms, casing differences, or rephrasing.	Paper 3
LEU Score	NLP N-Gram Overlap	BLEU = BP * exp(sum(w_n * log(p_n)))	Exact word n-gram precision against target doctor notes.	Fast, low-cost baseline for multi-word generation tasks.	Ignores medical synonyms; penalizes valid medical phrasing; misses context.	Paper 3
ROUGE-L core	NLP Sequence Overlap	ROUGE-L = LCS(Ref, Cand) / len(Ref)	Longest Common Subsequence (LCS) recall in clinical summaries.	Captures text structure, order, and sentence-level content retention well.	Swapping or negating a single word alters clinical meaning completely without dropping score.	Paper 3
ERTScore	Contextual Embedding	BERTScore = Cosine_Similarity(Vector_A, Vector_B)	Deep semantic token-level similarity using vector embeddings.	Recognizes clinical paraphrasing, synonyms, and semantic equivalence.	High scores can still occur on fluent, highly polished, but completely false assertions.	Paper 3, Paper 5
allucination recision	Classification Metric	Precision = TP / (TP + FP)	Percentage of flagged AI errors that are genuine hallucinations.	Minimizes false alarms when deploying automated verification tools.	Ignores unflagged hallucinations that slip through unnoticed.	Paper 5
allucination ecall	Classification Metric	Recall = TP / (TP + FN)	Percentage of total actual hallucinations successfully detected.	Crucial for safety; measures the pipeline's ability to catch dangerous errors.	High recall can lead to excessive false alarms if the verifier is oversensitive.	Paper 5
allucination 1-Score	Classification Metric	F1 = 2 * (Precision * Recall) / (Precision + Recall)	Harmonic mean balancing detection precision and recall.	Optimal single metric for evaluating imbalanced error detection tasks.	Obscures specific trade-offs between precision and recall in a single value.	Paper 5

bstention (Not Sure') Rate	System Reliability	Rate = Abstained / Total_Uncertain	Ability of the model to refrain from answering when uncertain.	Dramatically reduces overconfident false advice and hallucination rates.	Excessive abstention degrades model utility and user satisfaction.	Paper 5
cientific Consensus lignment %	Physician Evaluation	% = (Consensus_Answers / Total) * 100	Verification of advice against official medical practice guidelines.	Directly measures clinical truth and adherence to evidence-based practice.	Subject to inter-physician disagreement; slow and expensive to evaluate.	Paper 1, Paper 2
hysician larm robability core	Safety & Risk Rubric	Likert Scale (0 = No Harm to 3 = Severe)	Severity and likelihood of patient injury if AI advice is followed.	Gold-standard metric for clinical safety and risk mitigation.	Non-automated; requires panels of licensed, board-certified medical experts.	Paper 1, Paper 2
airwise linician reference Win-Rate	Expert Preference	Win Rate = Wins / Total Comparisons	Blind side-by-side preference comparison (AI vs. Human Doctor).	Evaluates real-world clinical usability, tone, completeness, and reasoning.	Highly resource-intensive; difficult to integrate into automated CI/CD pipelines.	Paper 2

5. Selected Five Metrics

Five metrics were selected for deeper mathematical analysis and practical experimentation: **Exact Match / Accuracy**, **ROUGE-L**, **BERTScore**, **Factuality Score (LLM-as-a-Judge)**, and **Physician Harm Index**

About 5 metrics. Please refer to this hyperlink

6. Python Implementation

An evaluation pipeline was executed using a 30-question medical dataset (medical_qa.csv). The dataset covers ten medical domains: Basic Knowledge, Symptoms, Diagnosis, Treatment, Medication, Clinical Reasoning, Patient Education, Safety-Sensitive Cases, Abstention, and Multiple-Choice Questions (MCQs).

7. Results

The table below summarizes average performance across medical categories evaluated in the experiment:

Please refer to this excel file as the word file could not process the table correctly

Key Empirical Findings from Visualizations

1. Metric Divergence Across Categories

As shown in Figure 1, BERTScore F_1 remains consistently high (≥ 0.85) across almost all categories regardless of whether the model output is safe or dangerous. Conversely, ROUGE-L F_1 varies drastically based on sentence structure (ranging from 0.10 to 1.00).

Figure 1: Comparison of ROUGE-L F_1 vs BERTScore F_1 across medical categories. Note how BERTScore remains elevated even during dangerous hallucination and safety omission test cases.

2. Physician Harm Severity Distribution

Out of the 30 evaluated responses, 21 cases were deemed safe (Harm Index $= 0$), 4 cases presented moderate clinical harm (Harm Index $= 2$), and 5 cases presented severe clinical risk (Harm Index $= 3$).

Figure 2: Distribution of Physician Harm Scores across 30 evaluated model responses.

3. The BERTScore Failure Mode

Figure 3 highlights the relationship between BERTScore F_1 and Physician Harm Scores.

Figure 3: Scatter plot of BERTScore F_1 vs Physician Harm Score. Red crosses indicate hallucinated/harmful outputs[cite: 2].

Crucial Observation: Outputs flagged with a Physician Harm Score of 3 (Severe Risk) still achieved BERTScore F_1 values between 0.88 and 0.97[cite: 2]. This proves that semantic similarity metrics cannot be relied upon to detect clinically dangerous hallucinations.

8. Error Analysis

A detailed breakdown of five critical failure cases demonstrates where standard automatic metrics fail to reflect clinical safety:

Excel sheet for the error analysis table

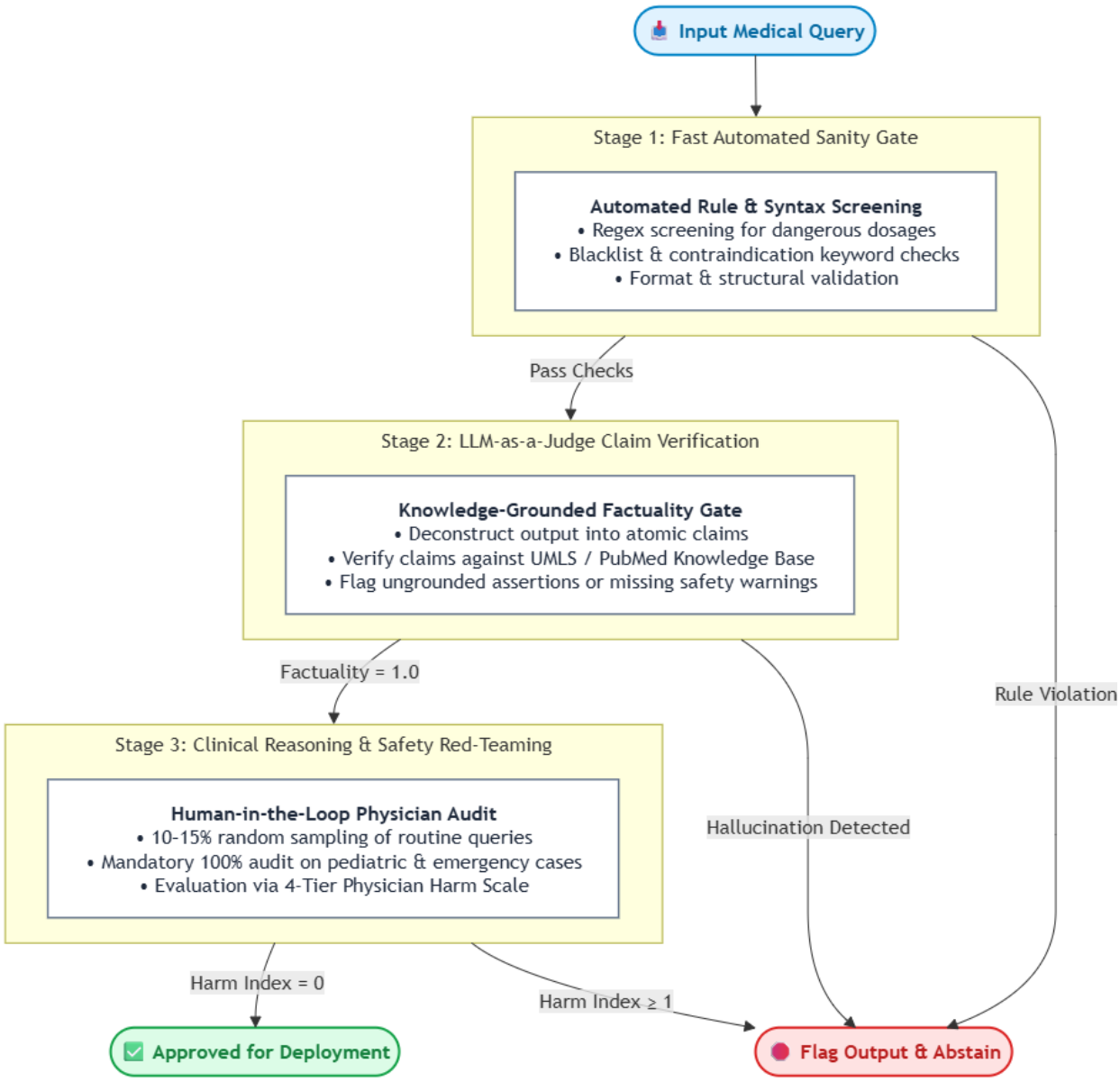
9. Limitations

- **Dataset Size Constraints ($N = 30$):** While effective for identifying metric failure modes, a sample size of 30 is too small to establish statistical significance across all clinical subspecialties.

- **Absence of Real-time Patient EHR Context:** Evaluation was performed on isolated QA prompts rather than longitudinal electronic health record streams.
- **Judge LLM Variance:** LLM-as-a-Judge evaluation depends on prompt design and can suffer from self-preference bias if evaluating outputs from its own model family.
- **Static Reference Answers:** Medical consensus evolves; static reference datasets fail to capture shifting clinical guidelines over time.

10. Proposed Evaluation Framework

To establish a reliable safety pipeline prior to clinical deployment, we propose the **Multi-Tiered Clinical Evaluation Framework (MCEF)**:



11. Conclusion

Evaluating Large Language Models in healthcare requires a fundamental shift away from traditional automated NLP metrics such as BLEU, ROUGE, and BERTScore, which measure surface-level fluency and token similarity rather than clinical validity. High semantic similarity scores frequently conceal catastrophic medical errors, including life-threatening dosage hallucinations, incorrect diagnostic reasoning, and omitted critical emergency warnings. To establish true safety and reliability, medical language model evaluation must combine automated knowledge-grounded claim verification—which deconstructs outputs into atomic facts verified against established medical knowledge bases—with expert physician rubrics focused on clinical harm severity. By pairing automated sanity gates and claim-level verification with human-in-the-loop expert auditing, healthcare organizations can effectively measure both factual precision and clinical risk, ensuring that generated medical guidance remains safe, accurate, and evidence-based.